



Twice considered: Charter schools and student achievement in Utah

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ABSTRACT

A relatively small state, Utah presents an interesting case to study charter schools given its friendly policy environment and its significant growth in charter school enrollment. Based on longitudinal student-level data from 2004 to 2009, this paper utilizes two approaches to evaluate the Utah charter school effectiveness: (a) hierarchical linear growth models with matched sample, and (b) general methods of moments with student-fixed effects regressions. Both methods yield consistent results that charter schools on average perform slightly worse as compared to traditional public schools, a result that is primarily affected by the low effectiveness and high student mobility of newly opened charter schools. Interestingly, when charter schools gain more experience they become as effective as traditional public schools, and in some cases more effective than traditional public schools. This research has implications for local and state charter school policies, particularly policies that avoid “start-up” costs associated with new charter schools.

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1. Introduction

The United States charter school movement has been driven by a multitude of arguments, including the expectation for market-based reforms to improve the quality of the current K-12 system of public education, the need for increased options for a dissatisfied customer base, and the ability of charter schools to educate children using fewer resources. Importantly, advocates predict that charter schools will benefit both students who actively choose their school—a direct effect—and non-choosers who remain behind in their assigned public schools—a systemic effect. Although both proposed benefits are equally important, this paper examines the direct effect of charter schools. Specifically, we address how charter schools affect the achievement of the students who attend them.

Since large-scale charter school systems were implemented in the 1990s, there has been a growing body of empirical research examining the relative effectiveness

of charter schools and traditional public schools (TPSs). Recent literature reviews on this issue, however, reveal the challenges of studying charter school effects. Among all the charter school achievement studies, only a small percentage of studies are considered methodologically rigorous, relying on either lottery-based or student-level growth-based methods (Betts & Tang, 2008, 2011). Depending on the student population served, operational years, and other factors, different studies found that charter schools have produced negative, positive, or no effect on student achievement (Betts & Tang, 2008, 2011; Miron, Evergreen, & Urschel, 2008; Teasley, 2009).

More recently, charter school studies using data nationwide or data from multiple states are also inconclusive (CREDO, 2009; Davis & Raymond, 2012; Gleason, Clark, Tuttle, & Dwyer, 2010; Lubienski & Lubienski, 2006; Tuttle, Gleason, & Clark, 2012; Zimmer, Gill, Booker, Lavertu, & Witte, 2012). Although national and multi-state studies provide a representative picture of charter schools in general, it is possible that the mixed results are the consequences of different policy environments. Charter school laws vary widely on mission, finance, regulation, and support from state to state. It is reasonable to expect different policies and implementation to influence

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charter school outcomes substantially. Consequently, the evidence of state-specific studies is important as it provides a more comprehensive understanding of the variations of charter school effectiveness in different policy environments. However, only a small number of states to date provide integrated longitudinal student level data for rigorous state-wide studies. By providing a specific state analysis, this study contributes to this body of research.

Based on longitudinal student-level data from 2004 to 2009 in Utah, this paper is one of the few studies to utilize two alternative methods, hierarchical linear growth models with matched samples and general methods of moments (GMM) with student fixed-effects models, to evaluate charter school effectiveness. In addition, with the availability of student residential information, the paper improves upon existing research in controlling for unobserved neighborhood characteristics when matching charter and TPS students. This study also presents a timely and interesting case with significant policy implications. Despite the overwhelming support of charter schools among politicians and increasing interest by parents, the information on Utah charter school effectiveness remains largely based on anecdotal evidence. As a state with both friendly charter school policies and the lowest per pupil funding, Utah presents a unique case.¹ Also, in comparison to charter schools in many other states and locales, Utah charter schools serve a predominately white and non-poverty student population. In that sense, this study extends existing literature and makes a potential contribution to generalizing the effects on charter schools as a whole.

2. The Utah charter school program

Among all the 40 states with charter school laws, the Utah charter law structure was ranked as the 4th strongest or most charter-friendly by the *Center for Education Reform* in 2010 and 11th in 2011.² Utah officially established charter schools in 1998. The Utah State Office of Education (USOE) Charter School Division describes the purpose of charter schools as:

Charters schools offer parents and students additional choices about where students attend school and the school's curricular emphasis. They allow educators freedom to try new strategies to inspire students and to experiment with innovative ways of educating students. Also, charter schools allow individuals and

organizations outside of the traditional education system to create and run public schools.³

The growth of the number of charter schools and the charter school enrollment in Utah has been steady. Charter schools in the state had 390 students in the first year of operation compared to about 33,000 ten years later in 2008–2009, which accounts for about 5% of all public school students in Utah. In 2008–2009, there were 65 operational charter schools.⁴ Currently, there are no caps on the number of charter schools. Following the 2010 Legislative Session, the State Board of Education was given the authority to remove the cap on charter school enrollment contingent on the availability of legislative appropriations (see Utah Code §53A-1a-502.5).

According to provisions of the Utah Code §53A-1a-506, Utah charter schools, as a public school choice option, must maintain an open enrollment similar to their traditional public school peers. Currently, charter schools in Utah have a higher proportion of elementary students and a lower proportion of high school students than TPSs. Policymakers and practitioners alike anticipate that the existing enrollment will lead to further demand for charter school availability as a student progresses from the elementary to secondary school level. Reportedly, many charter schools have waiting lists (Rorrer, Hausman, & Groth, 2006). When enrollment space is exceeded by demand, the charter school is required to use “random” selection.⁵

Utah charter schools may be authorized by local school boards, the Utah State Charter School Board with recommendation to the State School Board, and—as of 2010—higher education institutions. The majority of the charter schools are authorized by the State Board. According to Utah Administrative Code (R277-470-6), charter schools must be approved two years prior to opening their doors to students. Moreover, charter school applicants are required to attend an orientation and training session (Utah Administrative Code R277-470-4) provided by the State Charter School Board.⁶ Participation in the training may result in priority status for approval as well as additional funds.

Utah charter schools receive revenue from state funds (e.g., Local Revenue Replacement Program, Minimum School Fund, Revolving Loan Fund for capital outlay, School LAND Trust funds), federal funds (e.g., Federal Dissemination grants, Federal Start Up and Implementation Awards),

¹ Charter schools in Utah, as in any other states, cannot be affiliated with or restrict admissions based on a particular religion. However, frequently when discussing public education in Utah, questions arise with regard to the influence of religion in the state. Although an important question, it was not the focus of this inquiry. We only utilized official records maintained by the Utah State Office of Education, which does not contain any religious-related data on students.

² Please see <http://www.edreform.com/upload/ranking.chart.pdf> and <http://www.edreform.com/download/CER-Charter-Laws-2010.pdf>. The Center for Education Reform based their ranking on multiple factors, such as the existence of caps on the number of charter schools or enrollment, funding, start-up assistance, and the amount of regulation over charter school operations.

³ Please see <http://www.schools.utah.gov/charterschools/Frequently-Asked-Questions.aspx>.

⁴ Fourteen new charter schools have opened up between 2009–2010 and 2010–2011.

⁵ There are mandatory and optional preferences considered for the random selection. Mandatory preference refers to those schools where a TPS has converted to a charter school. Optional preference must be based on the school's charter and offers preference consistent with parent involvement with charter development, siblings attending schools, returning student status, district, municipality, and/or proximity of residence and parent is a licensed classroom teacher. <http://www.schools.utah.gov/charterschools/Frequently-Asked-Questions.aspx#6>.

⁶ The training topics include, but are not limited to, implementation, statutory, and Charter Board, financial, data management, legal, and funding requirements.

and other sources (e.g., grants, endowments, gifts, property donations). The Local Replacement Program and Revolving Loan Fund are specifically intended to reduce the disparity in revenue generating capacity between charter schools and TPSs, similar to the purpose of the federal funding sources for charter schools. Current state fiscal policies require districts to contribute a portion of the local property tax revenue they raise to the Local Replacement Program according to the number of charter students residing within their boundaries. Charter schools in Utah are not eligible for state-supported transportation funding.

3. Methodological challenges in charter school research

Estimating the effects of charter schools is methodologically challenging. The most serious problem arises because charter school students are self-selected and may be systematically different in many ways from those who remain in TPSs (Betts et al., 2006). For example, parents' choice to enroll their child in charter schools is based on factors such as parental preference, motivation, and their involvement in their child's education, which are usually unobserved. Ideally, the influence of these factors on student achievement needs to be controlled when estimating how the attendance of charter schools has influenced student achievement.

Several empirical strategies have emerged in the research to eliminate the self-selection bias, including lottery-based, matching, and value-added student fixed effect models (Betts & Tang, 2008, 2011). The lottery studies, generally regarded as the best approach in eliminating the bias, are to randomly assign students to charter schools and TPSs through admission lotteries, which enables researchers to compare achievement gains of lottery winners to losers (Abdulkadiroglu, Angrist, Dynarski, Kane, & Pathak, 2009; Gleason et al., 2010; Hoxby & Murarka, 2007; Hoxby & Rockoff, 2004; Tuttle, Gleason, & Clark, 2012). However, this experimental method is often infeasible, particularly when charter schools are not sufficiently oversubscribed, when admissions are not based on random selection, or when the information on students who did not get selected is unavailable. Even when lotteries are used, the results of lottery-based studies can only be generalized to a limited set of schools that are popular enough to have long waiting lists (Betts & Tang, 2008, 2011).

The matching approach reduces the self-selection problem by creating a matched sample of TPS comparison students that are as similar as possible to charter school students, and then statistically compare the achievement gains between the matched charter and TPS students. CREDO (2009) used the Virtual Control Record (VCR) technique to match each charter school student with a student enrolling in a TPS that have students who transferred to charter schools, based on multiple demographic variables and prior test scores. In other words, the VCR method produced pairs of students that have mirror images except that one is schooled in a charter school and the other in a TPS. Similar matching designs have been frequently used to control for selection bias (Davis & Raymond, 2012; Miron, Cullen, Applegate, & Farrell, 2007; Solmon, Paark,

& Garcia, 2001; Witte, Wolf, Cowen, Fleming, & Lucas-McLean, 2008). The matching approach can greatly reduce the self-selection bias, if the decision whether to enter a charter or a TPS of a given student is based on observed variables. However, it is possible that matching will not totally purge the selection bias, if some unobservable variables, such as motivation or parental involvement in education, determine whether a student enrolls in a charter school (Rosenbaum & Rubin, 1983).

Student fixed-effects models mitigate the self-selection bias by comparing students to themselves. Based on longitudinal student-level data that provide repeated observations of student achievement across school sectors, test score gains of students in charter schools are compared to the test score gains made by the same students while they were in TPSs, either before or after they were in a charter school. Any unobserved student characteristics related to both the likelihood to attend a charter school and achievement can be removed by the fixed-effects transformation, if these unobserved characteristics do not change over time.

With the increasing availability of longitudinal student level data, researchers have used the student fixed-effects method to study charter school achievement (Ballou, Teasley, & Zeidner, 2008; Betts, Tang, & Zau, 2007; Bifulco & Ladd, 2006; Booker, Gilpatric, Gronberg, & Jansen, 2007; Hanushek, Kain, Rivkin, & Branch, 2007; Imberman, 2011; Nicotera, Mendiburo, & Berends, 2011; Sass, 2006; Zimmer & Buddin, 2006; Zimmer et al., 2012). Some of these studies further refine their analysis by comparing charter schools by several features including operating years, educational levels, types of students served, pre-existing status, and their organization. In addition, the effect of student mobility on student learning as part of the charter school effect has often been examined further because student turnover rates are usually much higher compared to TPSs.

Although student fixed-effects models are effective in reducing student self-selection bias, there are several potential limitations of the method. First, it can only eliminate unobserved differences between charter and TPS students that remain constant over time. Second, all time-invariant variables such as students' gender and ethnicity are included in the term of the individual student fixed-effects and cannot be estimated. Finally, fixed-effects models rely only on "switchers"—students who have both attended charter schools and TPSs—by comparing achievement gains before and after they switched between charters and TPSs. If the switchers cannot represent all charter school students, then the external validity of the results will be significantly reduced. Indeed, this is likely the case in Utah, where more than one-third of students who enrolled in charter schools have never entered a TPS during their educational process.⁷ It is possible that

⁷ During 2004–2009, among all students who have attended any charter school for at least one year, about 43% never attended in TPSs. Half of these students, however, just started kindergarten or first grade in school year 2008–2009, as nine new charter schools opened that year. With the fast growth of charter schools, we expect to see more students start schooling in charter schools and remain in the charter school sector.

the “switchers” might have characteristics that are distinct from the students who never entered a TPS.

In sum, because the lottery-based method, the matching method, and the student fixed-effects models all suffer from different potential biases, studying charter schools using multiple methods to purge the confounding effects of self-selection in choice participation, such as this study, is imperative. Specifically, in this study we utilize both the matching and student fixed-effects approaches to estimate the effectiveness of Utah charter schools.

4. Data

The state's Data Warehouse since 2003–2004 is complete with a record of all students' history in Utah public schools, including their residence zip code, enrollment, attendance, demographic data for each year, and records of any standardized test taken starting in the first grade through twelfth grade. Students can be tracked over the years through a unique identifier, which retains their anonymity, as long as they remain in the Utah public school system. In addition, the state provides detailed information on every public school in Utah, including charter schools. The information includes but is not limited to the school type, location, grade span, and opening and closing years. Virtual charter and TPS schools are omitted from our analysis. Also omitted are the disciplinary facilities and schools that exclusively serve children with special needs.

The student achievement data are based on performance on state sponsored criterion referenced tests (CRT), which are usually administered in April of each academic year. CRTs provide information about the core skills and abilities students have acquired during the school year or course. The scores are reported on the same scale for all tests in math, language arts, and science, making student performance for different years and different tests comparable. The language arts and math CRT tests are administered in grades 2–11.⁸ Language arts tests are grade-specific, where students take the test that corresponds to the grade in which they are enrolled. The tests for mathematics for grades 1–6 and grades 7–11 are different, however. Math tests are grade specific in grades 1–6 and course specific in grade 7–11. Starting from grade 7, Utah students can take a variety of math courses such as math 7, applied math, pre-algebra, elementary algebra, algebra I, algebra II, and geometry. Students take a sequence of math courses in different grades based on their prior performance and teacher recommendations. As a result, students in the same grade can take CRT tests for different courses. The diverse course-taking patterns of students make it very hard to create longitudinal data to reflect math achievement gains that are comparable across students. Consequently, in this study, we only use elementary math scores in grades 1–6, which are consistent in core curriculum. For easier interpretation, all the scale scores are converted to standard scores with a mean of zero and

standard deviation (SD) of one, specific to grade, subject, and year.⁹

Different from many other states, Utah charter schools tend to serve more white and non-poverty students than TPSs. According to Table 1, charter schools have lower percentages of Hispanic students, low-income students, special education students, and English language learners (ELL) than TPSs. For example, although the average percentage of White students in TPSs dropped from 82.8% in 2004 to 78.1% in 2009 due to the demographic shift in the state, the average percentage of White students in charter schools remains high, consisting of 83.2% of all charter enrollment in 2009. Although charter schools now serve higher percentages of low-income and ELL students than they did in 2004, the proportions of these students remain significantly lower than TPS schools. Table 1 also shows that charter school students have slightly higher levels of performance on the state's language arts and math CRT tests. However, it is premature to conclude that charter schools are more effective since student test scores are highly correlated with student demographic characteristics.

5. Methods

Relying on longitudinal student-level data from 2004 to 2009, we use two different approaches to estimate the influence of charter school attendance on student achievement in Utah.¹⁰ The first approach utilizes hierarchical linear modeling (HLM) to compare students' achievement growth in charter schools and TPSs. With this method, we are able to include all students who attended charter schools for at least one year during 2004 and 2009 in the analysis. In order to mitigate the self-selection bias, we first use the matching techniques to create a comparison group of charter and TPS students that are as similar as possible to each other based on multiple variables. The matching techniques will result in a sample that includes not only the “switchers” but also the “stayers”—students who have remained in the charter schools or TPSs for all the years in the dataset.

Our second method utilizes student-fixed effects regressions that compare the achievement gains of the same students in charter schools and TPSs. This method is supposed to have greater internal validity than the longitudinal HLM growth models because it better controls for selection bias, as students are compared to themselves as they switch between charter schools and TPSs. The main disadvantage of the method, however, is that it focuses only on the “switchers” and ignores the “stayers.” In that sense, it has less external validity than the HLM growth models because the HLM models are able to include both the “switchers” and “stayers.”

Since each method is susceptible to different potential biases, we compare the results from both methods as a

⁸ Due to legislative changes, the CRTs are no longer administered in Grade one since 2008.

⁹ Although the math CRT test changed in both test contents and scales in 2009, after standardization of the scale scores by grade and year, there should not be much problem in interpretation.

¹⁰ As a result of the data availability, a lottery study is not possible at this time.

Table 1
Descriptive analysis for charter schools and traditional public schools, 2004 and 2009.

	2003–2004			2008–2009		
	Charter school (<i>n</i> = 2717)	TPS (<i>n</i> = 394,160)	Diff	Charter school (<i>n</i> = 23,977)	TPS (<i>n</i> = 443,239)	Diff
Total enrollment	0.68%	99.32%		5.13%	94.87%	
% Asian	1.73	1.63	0.1	1.94	1.73	0.21
% Black	1.73	1.18	0.55	1.59	1.48	0.11
% Caucasian	85.09	82.83	2.26	83.24	78.14	5.1
% Hispanic	7.29	11.06	−3.77	9.31	14.97	−5.66
% American Indian	2.69	1.63	1.06	0.92	1.56	−0.64
% Pacifica islander	1.36	1.41	−0.05	1.38	1.61	−0.23
% unidentified	0.11	0.25	−0.14	1.63	0.52	1.11
% FRL	16.08	34.15	−18.07	22.64	35.10	−12.46
% ELL	0.77	6.03	−5.26	1.32	6.82	−5.50
% special education	8.17	12.48	−4.31	11.12	12.40	−1.28
Math scaled score	166.06	164.99	1.07	164.51	163.07	1.44
Language arts scaled score	167.63	166.50	1.13	168.53	167.27	1.27

The core and the scale for math CRT has changed in 2009, so it is inappropriate to compare math CRT scores in 2009 with those in 2004.

robustness check. In addition, because the two methods use different samples, it is possible that the differences in estimates are not due to the approaches rather than the different samples. To check this possibility, we also estimated the HLM growth models focusing solely on the “switchers” as a sensitivity analysis.

In both methods, we control for two confounding factors that have been found in the literature to influence student performance, charter school experience and student mobility. First, literature has shown that charter school performance generally improves when they gain more experience. Traditional public schools, on the other hand, do not seem to experience the same performance growth, in part because of the presence of necessary support and resources when new TPSs are open (Sass, 2006). By 2009, 30 Utah charter schools had no more than 3 years of experience, 9 were in their 4th year of operation, 11 in their 5th year, and the remaining 15 charter schools had been open for 6 or more years. To capture the possible vintage effect of charter schools, we follow the literature and categorize charter schools by their operational years through a series of dichotomous variables (Bifulco & Ladd, 2006; Booker et al., 2007; Hanushek et al., 2007; Sass, 2006).

Second, research has shown that student mobility is usually disruptive in student learning (Cowen, Fleming, Witte, & Wolf, 2012; Lash & Kirkpatrick, 1990, 1994). In addition, transfers between charter schools and TPSs might have very different effects on student achievement from the transfers within each sector, because of the institutional and organization differences between the two types of schools. As shown in Table 2, Utah charter schools have lower student turnover rates than TPSs, unlike the case in some other states (Bifulco & Ladd, 2006; Booker et al., 2007). However, charter school students are much more likely to transfer back to a TPS than to another charter school, which might reflect the fact that there are not many charter schools close by to choose from. TPS students, on the other hand, also tend to transfer between TPSs instead of moving to a charter school. Utah maintains various open enrollment opportunities, through which students are free to transfer to another TPS either within or outside the district boundary provided the receiving school has space. Table 2 also shows that charter schools have higher percentages of students exiting the Utah public school system than TPSs. That is, these students do not appear in the Data Warehouse records due to reasons such as moving out of

Table 2
Student mobility in charter school and traditional public schools, between 2007–2008 and 2008–2009.

Grade	From charter school			From TPS		
	To another charter school	To TPS	Out of Utah public school system	To another TPS	To charter school	Out of Utah public school system
1	1.4	9.2	7.1	13.6	1.3	5.7
2	1.8	8.4	5.9	14.6	1.3	5.3
3	1.0	9.4	6.2	13.2	1.2	5.0
4	2.3	10.8	6.7	12.5	1.4	5.0
5	1.8	11.6	7.1	25.7	1.3	5.8
6	5.1	25.9	10	75.5	1.4	4.8
7	5.1	13.6	7.3	16.6	0.9	4.5
8	19	26.3	15.5	22.4	1.8	4.8
9	7.0	30.8	12.4	72.6	1.5	5.4
10	1.9	17.2	11.7	9.0	1.0	6.8
11	0.5	12.5	14.5	8.1	0.5	10.2
Total	3.8	15	8.8	25.3	1.3	5.7

Table 3

Non-structural and structural move between charter schools and TPSs, between 2007–2008 and 2008–2009.

Grade	Structural move (%)		Non-structural move (%)	
	Charter schools	TPSs	Charter schools	TPSs
1	0.0	0.4	10.6	14.5
2	0.0	1.8	10.2	14.0
3	0.5	0.7	9.9	13.7
4	0.0	0.5	13.1	13.4
5	2.2	17.8	11.2	9.2
6	11.6	73.5	19.5	3.3
7	0.0	11.0	18.8	6.5
8	29.3	15.1	16.0	9.1
9	20.4	69.3	17.4	4.8
10	0.3	0.0	18.7	10.0
11	12.9	8.7	0.0	0.0
Total	5.6	17.3	13.1	9.2

the state, dropping out of school, attending a private school, or being home schooled.

Following the literature, we classify student mobility into two broad categories: structural move and non-structural move. Structural move is defined as a student in the highest grade offered, or terminal grade, in a school and moves into a new school the following year. Structural moves are common as students move from elementary to junior high to high school. Non-structural move means students who are not in a terminal grade change their schools from year to year. This includes, for instance, family relocation, inter-district choice, intra-district choice, transfer between charter and TPSs, or transfer among charters. As research has shown, structural moves tend to have a larger disruptive effect than non-structural move, perhaps due to very different curriculums and expectations (Booker et al., 2007).

As Table 3 shows, while Utah charter schools have higher non-structural mobility than TPSs, their average structural mobility is only one-third of that of TPSs. A closer examination of mobility by grade level reveals that TPSs have higher structural mobility rates in certain grades because of the Utah TPSs grade span configurations. The highest structural mobility occurs in the elementary to middle school transition (grade 5 to 6 or grade 6 to 7) and middle school to high school transition (grade 8 to 9 or grade 9 to grade 10). On the other hand, charter schools have more varied grade level combinations such as k-8, k-9, 7–12, or k-12. So their transition patterns across grades are very different from TPSs.

5.1. HLM with matched samples of charter school and TPS students

Although there are 41 school districts in Utah, charter schools attract the majority of their students from only several districts largely along the Wasatch Front. In addition, students in charters and TPSs are different, which partially accounts for the difference in achievement levels and growths between charter and TPS students. In order to make a more accurate estimation of charter school effects, we first created a comparable group of charter and TPS students through the VCR approach (CREDO, 2009; Davis &

Raymond, 2012). Then, based on the matched sample, we used two-level HLM growth models to analyze how charter school attendance influences the trajectories of student achievement growth (Raudenbush & Bryk, 2002).

5.1.1. Matching

To create a reliable comparison group for charter school students, we aimed to match each student who had attended a charter school at least once during the 2004 and 2009 time period with a TPS student who closely mirrors the charter school student on multiple variables. This is done through several steps. First, we identified all students attending a charter school in 2005 and tracked each of them back to 2004 and identified his or her resident zip code. All TPS students residing in the same zip code in 2004, including those who might have transferred to a charter school in later years, became potential matches for this particular charter school student.¹¹ Then, we limited the pool of potential matches to TPS students with exactly the same characteristics of the charter school student on grade-level, gender, race/ethnicity, eligibility for free/reduced lunch (FRL),¹² ELL status, and special education status. Then, among these potential matches, we identified the one whose achievement in 2004 was the closest to that of the charter school student.¹³ This procedure produced a matched sample of TPS students for all charter students in 2005. Once matched, these students were removed from the dataset so that they would no longer be matched again.

Next, among the remaining students in the dataset, we identified all students who first appeared in charter schools in 2006 and tracked them back to 2005. We repeated the steps above to match them with TPS students according to their demographic characteristic and prior achievement in 2005. We repeated the same procedure for all the following years up to 2009 until each charter school student was matched with a TPS student who has exactly the same demographics and the most similar baseline performance. Finally, based on the unique student identifier, we extracted all the information for each student in the matched sample to form a panel data from 2004 to 2009, or all the years that the students were in the Utah public school system, for the subsequent HLM growth model analysis.

The matching greatly reduced the selection bias because of the observable similarities between these charter school students and TPS students. Further, since the matched pairs are from the same resident zip code, it helps to mitigate a

¹¹ In CREDO's report (2009), the authors identified potential matches of charter school students as all students in their "feeder schools," or the TPSs that have students transferring to a given charter school. This method will not be appropriate here, because a substantial portion of Utah charter school students have never entered a TPS. For these students, they do not have a TPS as their feeder school.

¹² Data on eligibility for free/reduced lunch are collected for all students in Utah, including students in charter schools that do NOT offer a federally supported school meals program.

¹³ Students in grades one to six are matched based on scores of both math and language arts. Students in seventh grade and above are only matched based on language arts scores. Grade one students don't have prior scores, so they are matched when they get to grade two.

significant portion of selection bias since families residing in the same areas tend to be similar in unobserved ways and also should have similar knowledge about the availability of charter schools in the neighborhood (Witte et al., 2008). One potential drawback of our matching method using baseline test scores, however, is to reduce possible effect charter schools have for students in early grades, since test scores are unavailable for students until they reach the end of first grade.

5.1.2. HLM growth models with matched samples

Based on the matched sample of charter and TPS students, we estimated a series of two-levels of HLM growth models that compare student achievement growth to test whether charter school attendance produced better or worse achievement growth for students than they would have obtained in TPSs.

Level-1

$$Y_{ti} = \pi_{0i} + \pi_{1i}(\text{grade})_{ti} + \pi_{2i}(\text{charter})_{ti} + \pi_{3i}(\text{other time-varying variables})_{ti} + e_{ti}$$

Level-2

$$\pi_{0i} = \beta_{00} + \beta_{01}(\text{student characteristic})_i + r_{0i}$$

$$\pi_{1i} = \beta_{10} + \beta_{11}(\text{student characteristic})_i + r_{1i}$$

$$\pi_{2i} = \beta_{20} + \beta_{21}(\text{student characteristic})_i + r_{2i}$$

$$\pi_{3i} = \beta_{30} + r_{3i}$$

The level-1 model is a within-student repeated-observations model, where student i 's achievement of math and language arts at time t , measured as Y_{ti} , is represented by an individual linear growth trajectory. Grade $_{ti}$ is the grade at time t for student i , and π_{1i} is the growth rate of student achievement as they progress through grades. The variable of interest is the time-varying dichotomous variable, charter $_{ti}$, indicating whether student i attended a charter school in year t . The coefficient π_{2i} represents the effect of charter school attendance of a student in a particular year, or the deflection from the expected achievement growth. Charter schools are further categorized by years in operation, which capture the possible vintage effect. Other control time-varying variables at level-1 include FRL, student mobility, and programmatic characteristics such as enrollment in special education or English language programs. All of these variables might change over time for a particular student. In addition, student achievement growth varies across individuals, depending on their family backgrounds, school input, and peer environments. Since students tend to transfer between schools from year to year, school characteristics are time-variant as well. We include a full set of student composition variables of the

school a student attended in a particular year to control for the peer effects in order to isolate the effect of attending charters schools.¹⁴ The level-1 model also includes a set of grade dummies and year dummies to control for the test fluctuations pertaining to grade and year.

The unit of the level-2 model is student. All the time-invariant student characteristics that affect student achievement and growth, such as race/ethnicity and gender, are included in the level-2 models to predict their mean achievement, the rate of achievement growth, and the differential effects of charter schools on their achievement.

5.2. Value-added student fixed-effects models

In our second approach of estimating the effectiveness of charter schools, we rely on repeated observations on individuals to control for student fixed effects by comparing test score gains of students in charter schools to the gains of the same students when they attend TPSs. In other words, we focus only on “switchers.” Typically, researchers follow a general cumulative model of student learning, where student achievement is a function of entire input histories of family inputs and school inputs including their attendance in charter schools, and students' innate ability. The model allows the effects of inputs from previous years on current student achievement to diminish at an annual depreciation rate of $1 - \theta$, where $0 < \theta < 1$.¹⁵ The general model for the achievement of student i in grade G in year T can be expressed as Eq. (1) (Bifulco & Ladd, 2006; Hanushek et al., 2007; Sass, 2006):

$$Y_{iGT} = \alpha \sum_{t=1}^T \theta^{T-t} CH_{igt} + \beta \sum_{t=1}^T \theta^{T-t} \mathbf{X}_{igt} + \delta \sum_{t=1}^T \theta^{T-t} \mathbf{S}_{igt} + \sum_{g=1}^G \theta^{G-g} \gamma_{ig} + \varphi_{GT} + v_{iGT} \quad (1)$$

where CH_{igt} is a dichotomous variable indicating whether student i for grade g attends a charter school in year t . The summation of the term from time period 1 to the current time period T indicates that the effect of charter school attendance on current achievement accumulates over time, with the attendance in the past having a weaker influence (θ^{T-t}) than the attendance in the current year. Similarly, $\mathbf{X}_{igt}$ is a vector of time-varying student characteristics including student mobility that have cumulative effects on student achievement. The sum of $\mathbf{S}_{igt}$ is the quality of schooling including school input and peer composition from all previous years. The term γ_{ig} is student systematic characteristics such as parental care and early family input, innate ability, and other unobserved factors. In addition, φ_{GT} is a

¹⁴ Since we explicitly model student transfer between schools across the years, it is not possible to use a 3-level HLM where individual students are nested within schools. In addition, cross-classification models between students and schools are not feasible because school characteristics, including student composition, change over time.

¹⁵ This means, for example, the school input in a student's fourth grade has a smaller effect on her fifth grade achievement than on her fourth grade achievement and an even lesser effect on her seventh grade achievement.

grade-by-year fixed effect that captures fluctuations in test performance across grades and cohorts that may introduce spurious correlation between student composition and achievement. v_{iGT} is the idiosyncratic error term that changes across time for each student.

To simplify Eq. (1), we rewrite the equation for $Y_{i(G-1)(T-1)}$, achievement of student i in grade $G-1$ and year $T-1$. We then take the difference between Eq. (1) for Y_{iGT} and for $\theta Y_{i(G-1)(T-1)}$ and get Eq. (2):

$$Y_{iGT} = \theta Y_{i(G-1)(T-1)} + \alpha CH_{iGT} + \beta X_{iGT} + \delta S_{iGT} + \gamma_{iG} + \tau_{GT} + \varepsilon_{iGT} \quad (2)$$

where $\tau_{GT} = \varphi_{GT} - \theta \varphi_{(G-1)(T-1)}$ and $\varepsilon_{iGT} = v_{iGT} - v_{i(G-1)(T-1)}$. Eq. (2) can be interpreted as a non-restricted value-added specification, where the current achievement level depends on lagged achievement that depreciates at a certain rate.¹⁶ Now that the equation includes the lagged dependent variable, the usual fixed-effects estimations will be biased due to autocorrelation and the correlation of the lagged variable and the error term. To solve this problem, the general methods of moments (GMM) with instrumental variables can be applied, where the student fixed effect, γ_{iG} , can be removed by first-differencing the data:

$$\Delta Y_{iGT} = \theta \Delta Y_{i(G-1)(T-1)} + \alpha \Delta CH_{iGT} + \beta \Delta X_{iGT} + \delta \Delta S_{iGT} + \Delta \tau_{GT} + \Delta \varepsilon_{iGT} \quad (3)$$

After first differencing, $\Delta Y_{i(G-1)(T-1)}$ is necessarily correlated with the error term $\Delta \varepsilon_{iGT}$ because $Y_{i(G-1)(T-1)}$ and $\varepsilon_{i(G-1)(T-1)}$ are correlated. To circumvent this problem, we follow the Arellano–Bond dynamic panel GMM estimator with fixed-effects using the second and deeper lags of student achievement as instrumental variables for $\Delta Y_{i(G-1)(T-1)}$ (Sass, 2006; Wooldridge, 2002). The idea is that two or more lags of Y are correlated with the lagged achievement $\Delta Y_{i(G-1)(T-1)}$, but are uncorrelated with the error term $\Delta \varepsilon_{iGT}$. Note that since the data are first differenced and in need of the second lag as the instrumental variable, only students with three consecutive years of data are included in the models.

6. The effect of charter schools on student achievement

6.1. Longitudinal HLM growth models results

The VCR matching procedures resulted in a comparison sample of 19,452 students who were observed at least once in a charter school during 2004–2009 and their TPS counterparts. Table 4 presents the baseline information of the

Table 4

Baseline characteristics of matched charter school students and TPS students.

	Charter school	Traditional public schools
Total enrollment	19,452	19,452
Asian	1.09	1.09
Black	1.04	1.04
Caucasian	88.06	88.06
Hispanic	7.92	7.92
Indian	0.41	0.41
Pacific islander	1.14	1.14
Unidentified	0.34	0.34
% FRL	24.45	24.45
% ELL	1.68	1.68
% special education	10.05	10.05
Math scaled score	167.48	167.67
Language arts scaled score	168.25	168.34

All the characteristics are based on the information from the year before a charter school student first appeared in the database and her match.

matched sample the year before a charter school student was matched. The pairs have exactly the same demographic characteristics. The starting math and language arts test scores are very close and all the differences between the two groups are not statistically significant.

A series of two-level HLM linear growth models were then performed based on the matched samples of charter and TPS students. The results from the unconditional models with no controls at either level, though not reported here, show that the proportion of the variance of achievement at level-2 is around 66–68%, depending on the subject, indicating substantial variation in average achievement levels among individual students.

Table 5 shows the estimate results of the conditional models with student and school control variables at both level-1 and level-2. The same analysis was repeated for elementary math, elementary language arts, and secondary language arts. As indicated earlier, secondary math was not estimated because the tests are course-based, and the scores are not comparable across grades. For each subject, four models were estimated. Model 1 only includes the dichotomous variable at level-1 indicating charter school attendance of a student in a particular grade. Model 2 adds the full set of student and school controls, including student mobility, which is described through two variables, move and structural move. Model 3 further distinguishes the operational years of charter schools through a set of dichotomous variables, while Model 4 categorizes student mobility into moves within and between the charter school and TPS sectors.

As the results of Column 1 in Table 5 indicate, students who attend charter schools have lower math test score gains by .067 SDs in elementary schools. After controlling for student and school variables as well as student mobility, the effect becomes –.059 SDs. The negative effect is primarily driven by newly established charter schools, as indicated by the results in Column 3. On average, students attending charter schools within their first year of operation experience about .15 SDs smaller gains in math than the students attending TPSs. The magnitude of the negative results decrease as charter schools gain more

¹⁶ If we restrict the value of θ to 1, the equation becomes a restricted value-added model and can be estimated through fixed-effects models that remove the student fixed effect term γ_{iG} . However, as shown in our later analysis (Table 6), θ is estimated to be .03 for elementary math, .06 in elementary school, and .12 in secondary schools. These results indicate that the assumption of no decay of the effect of prior input is false (Sass, 2006). Therefore, the restricted value-added fixed-effects models are not estimated in this study.

Table 5

Estimated effects of charter school attendance on student achievement: Hierarchical linear growth models based on matched samples.

	Math grades 1–6				Language arts grades 1–6				Language arts grades 7–11			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Growth rate	.024** (.006)	.032** (.006)	.03** (.006)	.03** (.006)	.005 (.006)	.008 (.006)	.007 (.006)	.007 (.006)	–.042** (.004)	–.044** (.004)	–.044** (.004)	–.044** (.004)
Charter school	–.067** (.006)	–.059** (.007)			–.079** (.006)	–.093** (.007)			–.008 (.008)	–.005 (.009)		
1st year			–.15** (.009)	–.117** (.012)			–.14** (.009)	–.132** (.012)			–.062** (.016)	–.036 (.02)
2nd year			–.078** (.009)	–.058** (.01)			–.106** (.009)	–.101** (.009)			–.01 (.014)	.012 (.016)
3rd year			–.032** (.01)	–.013 (.01)			–.082** (.009)	–.077** (.01)			–.018 (.013)	.001 (.015)
4th year			.01 (.012)	.019 (.012)			–.061** (.011)	–.056** (.012)			.002 (.015)	.019 (.016)
5th year			.107** (.014)	.127** (.015)			.02 (.014)	.026 (.014)			.066** (.018)	.084** (.019)
6th year/more			.081** (.016)	.10** (.016)			–.008 (.015)	–.002 (.015)			.03 (.018)	.046* (.019)
Mover		–.054** (.005)	–.029** (.006)			–.06** (.005)	–.011* (.005)			–.061** (.009)	–.055** (.009)	
TPS-to-TPS mover				–.03** (.007)				–.013 (.007)				–.063** (.011)
TPS-to-charter mover				–.064** (.01)				–.018 (.01)				–.086** (.014)
Charter-to-TPS mover				.055** (.016)				.017 (.015)				–.015 (.014)
Charter-to-charter mover				–.01 (.024)				–.014 (.023)				–.024 (.024)
Structural mover		–.102** (.017)	–.113** (.017)	–.106** (.018)		–.024** (.005)	–.059** (.017)	–.056** (.017)		–.019 (.01)	–.017 (.01)	–.025* (.011)
Student controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
School controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Deviance	353,958	339,446	339,235	339,208	353,912	336,220	336,145	336,151	131,635	122,645	122,636	122,619
# obs.		130,371					130,375				50,093	
# students		35,003					34,997				18,971	

A set of grade dummies and year dummies are included in the level-1 models across all specifications. Other control variables in the level-1 model in all specifications also include time-varying student characteristics (FRL, special education, ELL, absent days) and school characteristics (enrollment, % Black, % Hispanic, % Asian, % FRL students, % ELL, % special education students, and % drop outs). The time invariant variables in the level-2 model include: gender and a vector of race dummies. These results are not reported here due to space limit. Robust standard errors are reported in the parentheses.

* Statistically significant at .05 level.

** Statistically significant at .01 level.

experience. Starting from the fifth operational year, the trend is reversed. Students enrolling in charter schools with five and more years of experience obtain greater math gains than students attending TPSs. This confirms the findings from other studies that charter schools do learn from their experience (Booker et al., 2007; Sass, 2006). After experiencing significant start-up difficulties, their student performances improve once they are more established.

Student mobility has a negative effect on elementary math achievement. On average, a student transfer is associated with .03–.05 SDs decrease in their math test score gains, as indicated in Columns 2 and 3 of Table 5. The structural movement has an extra disruptive effect around .11 SDs on math score gains. When we further categorize student mobility as charter to TPS moves, charter to charter moves, TPS to TPS moves, and TPS to charter moves, we find different types of mobility have different effects on student achievement, as shown in Column 4 of Table 5. As expected, both moves from TPSs to charters and between TPSs have negative effects on student achievement. While transferring from one charter school to another seems to have no significant impact on student achievement, the transitions from charters to TPSs actually have a positive effect on student achievement. Although this seems to be odd at first sight, it is consistent with the findings in previous studies (Booker et al., 2007). Given the scope of this study, however, we cannot attribute an underlying reason for this.

After controlling for the different types of student transfers, the charter school effect on elementary math becomes less negative in their earlier years and more positive when they are more established. For example, the estimated negative effect of first-year charter schools on elementary math becomes smaller, changing from –.15 SDs in Column 3 of Table 5 to –.12 SDs in Column 4, and the magnitudes of the positive effects of more established charter schools increase. Note that the total effect of charter schools is a combination of the effects of charter attendance and student mobility. Although charter schools in Utah have lower mobility rates than TPSs, student mobility seems to have a more disruptive effect for students in charter schools than those in TPSs. The reason for this effect is not certain based on this study. However, this may indicate that charter schools have fewer structures in place (e.g., student transition plans) than TPSs to make newly transferred-in students better acclimated with their school and learning environment.

Columns 5–8 of Table 5 present the estimates of the impact of charter schools on elementary language arts achievement gains. Similar to the results of the math models, charter attendance on average has a negative effect on language arts test scores gains. The effect of attending a charter school that has opened no more than four years remains negative, although the magnitude of the negative effect decreases over time. Only as charter schools reach their fifth year of operation do they perform as well as TPSs. The effects of student mobility on language arts performance are very similar to its effects on math scores, but to a lesser extent.

The charter school effect on secondary language arts is somewhat different. As shown in Columns 9–12 of Table 5, secondary charter schools in general perform as well as

secondary TPSs. Thus, they do not seem to exhibit the same start-up difficulties as elementary charter schools. Only in their first year of operation do charter schools have a negative effect on students' secondary language arts performance, which lessens after controlling for different types of student mobility. In their second to fourth year of operation, their students perform at the same levels as TPS students. Starting from the fifth year of operation they actually perform better than TPSs.

6.2. GMM with fixed-effects results

Table 6 presents the estimates of the charter school effects on student achievement in elementary math, elementary language arts, and secondary language arts through the value-added dynamic panel GMM estimators, where the lagged dependent variable is included as a control variable. Similar to the HLM estimations, four models were estimated for each subject.

As Column 1 shows, charter school attendance has a negative effect on elementary math gains. Math achievement in the average elementary charter school is .11 SDs lower than the average elementary TPS. After controlling for student and school characteristics and student mobility, the negative effect lessens. The results are comparable to those of the HLM math models, as shown in Columns 1 and 2 of Table 5. In addition, consistent with the HLM results, the negative effect of charter schools is primarily driven by newly established charter schools, as indicated in Columns 3 and 4 in Table 6. The student math achievement in first-year charter schools is .14 or .12 SDs lower than students in TPSs, depending on the model specification. When charter schools are more established, the effect becomes less negative and when charter schools reach their fifth year of operation, they tend to perform better than TPSs.

Again, the structural movement of students is associated with negative achievement changes, a result similar to the HLM results. On the other hand, non-structural movement has shown small positive effects on student achievement, which is the opposite of the HLM results. When further categorizing student mobility, however, we find that most of the transfers have effects that are consistent with the HLM results. The only difference is that while transitions between TPSs are estimated to have a negative effect on student achievement in the HLM models, they seem to have a positive effect in the GMM models. Nevertheless, after controlling for student mobility, the charter school effect becomes less negative in their start-up years and more positive in later years, as indicated in Columns 3 and 4. This further suggests that charter schools are less effective than TPSs in accommodating the newly transferred-in students in learning.

Columns 5–12 present the estimates of the impact of charter schools on elementary and secondary language arts achievement. Similar to the HLM results, while elementary charter school students experience lower achievement gains in language arts than TPS students, the secondary charter school students in general gain as much as TPS students. The trends of the performance gains associated with years in operation are also similar to the HLM results. The longer a charter school exists, the better its students

Table 6

Estimated effects of charter school attendance on student achievement, Dynamic panel GMM with fixed-effects models based on “switchers”.

	Math grades 1–6				Language arts grades 1–6				Language arts grades 7–11			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Achievement _{t-1}	.029** (.003)	.027** (.003)	.028** (.003)	.028** (.003)	.059** (.004)	.062** (.003)	.057** (.003)	.057** (.003)	.108** (.006)	.117** (.006)	.117** (.006)	.117** (.006)
Charter school	–.111** (.009)	–.085** (.010)			–.107** (.009)	–.099** (.01)			–.023 (.013)	.019 (.013)		
1st year charter			–.139** (.012)	–.120** (.021)			–.128** (.012)	–.136** (.021)			–.001 (.022)	–.016 (.032)
2nd year charter			–.080** (.012)	–.062** (.014)			–.095** (.012)	–.091** (.014)			–.004 (.017)	–.008 (.022)
3rd year charter			–.045** (.013)	–.027** (.014)			–.081** (.013)	–.080** (.014)			–.018 (.019)	–.04 (.023)
4th year charter			–.027 (.015)	–.010 (.017)			–.078** (.015)	–.071** (.017)			.03 (.02)	.01 (.023)
5th year charter			.041* (.019)	.058** (.020)			–.02 (.019)	–.021 (.02)			.087** (.026)	.065* (.028)
6th year and more charter			.069** (.023)	.087** (.024)			.004 (.024)	.005 (.025)			.079** (.029)	.059 (.03)
Mover		.004 (.003)	.007 (.004)			.01** (.003)	.012** (.003)			–.007 (.005)	–.007 (.005)	
TPS-to-TPS mover				.006* (.003)				.011** (.003)				–.008 (.005)
TPS-to-charter mover				–.002 (.014)				–.021 (.014)				–.017 (.017)
Charter-to-TPS mover				.054** (.017)				.027 (.017)				.017 (.018)
Charter-to-charter mover				.006 (.02)				.017 (.028)				–.06 (.035)
Structural mover		–.056** (.005)	–.058** (.005)	–.058** (.005)		–.038** (.005)	–.041** (.005)	–.04** (.005)		–.02** (.005)	–.02** (.005)	–.019** (.005)
Student controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
School controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
# obs.		489,227				489,766				325,336		
# students		230,444				230,807				177,136		

All models include year/grade fixed effects. Student controls include FRL, special education, ELL, absent days. School controls include enrollment, % Black, % Hispanic, % Asian, % FRL, % ELL, and % special education students in school. These results are not reported here due to space limit. Time-invariant variables, such as gender and race, are not included because they do not change over time and cannot be estimated through the fixed-effects models. Robust standard errors are in parentheses.

* Statistically significant at .05 level.

** Statistically significant at .01 level.

perform in language arts. It takes five years for elementary charter schools to perform as well as TPSs in language arts, while the newly established secondary charter schools perform about the same level as TPSs. When secondary schools gain more experience after about four years, they actually perform better than TPSs in language arts. As with the HLM models, structural mobility of students has a negative effect on language arts test score gains. However, the estimated effects of non-structural student mobility are less consistent and mostly insignificant when further distinguished by the type of mobility. Once the student mobility is controlled, charter schools seem to demonstrate less negative effects on elementary language arts and no significant difference on secondary language arts.

6.3. Robustness check

So far, we have estimated the charter school effect with both the longitudinal HLM growth models and GMM regressions. The results have shown that, although the magnitudes of the estimated effect vary slightly in different methods and specifications; the directions and the trends of the effect remain consistent. One concern remains, however, since two methods focus on different samples. The HLM estimations focused on a matched sample of charter school students and TPS students, which includes both “stayers” and “switchers”, while the GMM estimators focused only on “switchers.” One way to make an apples-to-apples comparison of the two methods is to use the same set of students as a sensitivity analysis. Unfortunately, it is not possible to use the GMM estimators on the matched sample because the charter attendance of the “stayers” is a time-invariant variable and will drop out in the GMM estimators after first-differencing. However, it is possible to use the HLM models as an alternative approach to estimate the charter school effect on the “switchers”, the same set of students used for the GMM estimators.

Table 7 shows the results of the HLM models that were estimated based only on the “switchers,” which are reasonably consistent with the GMM results. Charter schools on average perform slightly worse than TPSs in elementary math and language arts and about the same in secondary language arts. Newly established charter schools tend to experience significantly lower performance gains than TPSs. As they obtain more experience after about four years, they actually outperform TPSs. The only result that is less consistent with the GMM analysis is that first year secondary charter schools exhibit some negative effect on language arts.

As noted earlier, the HLM growth models have better external validity than the GMM models because all students, both “switchers” and “stayers”, were included in the analysis. Since the HLMs focused only on the “switchers” and produced results relatively consistent with the GMM models on the same sample, we are reasonably confident that the HLMs in this study also achieved high internal validity. The remaining potential bias may not pose any serious problems and the estimated charter school effect is likely to be accurate. However, we cannot completely exclude the possibility that the direction and the magnitudes of the bias are similar for both methods.

7. Discussion

As charter school programs vary widely across the states, it is important to evaluate their effectiveness in different settings. Utah provides an interesting case to study, as there has been significant growth in both the number of and the enrollment in charter schools since their emergence as a public school choice alternative in the state in 1998. Numerous factors contribute to this growth in Utah. First, current statute does not impose caps on the number of or enrollment in Utah charter schools. Second, as most charters began as elementary schools, the demand for middle and high schools continues to require subsequent grade-levels for families who have selected them as their public school alternative. Despite their growth in the state, Utah charter schools enroll a lower proportion of students of color and students from low-income families than TPSs.

As charter schools are advocated generally as a means to improve student achievement, it is interesting to examine their effectiveness in a unique setting such as Utah. This paper is the first step to understand the effect of charter schools on student achievement in Utah. Utilizing longitudinal student-level data, we are able to use two different methods, HLM growth models with matched samples and GMM regressions with fixed-effects, to compare student achievement gains in charter schools and TPSs. As an additional sensitivity analysis, we also preformed the HLM growth models with the same set of students used for the GMM estimators. The results from different estimations are similar, indicating both HLM growth models and GMM regressions have similar levels of internal validity, while the HLM growth models have greater external validity.

Our analysis shows that Utah charter schools on average perform slightly lower than TPSs in elementary math and language arts and about the same in secondary language arts. Importantly, the negative results are mainly driven by charter schools in their organizational infancy. Among the 65 charter schools included in the analysis, almost half had been open for no more than three years. These schools may experience significant start-up difficulties (e.g., staffing, resources, and materials) and are less effective in cultivating integrated learning environments for students. As they mature, however, elementary charter schools perform similar to TPSs in language arts while better in math. Newly established secondary charter schools, on the other hand, perform as well as TPSs from their inception. After gaining experience through the years, secondary charter schools generally perform slightly better than secondary TPSs in language arts. The magnitudes of all effects are relatively small, however, mostly within .10 SDs.

Our finding that charter schools tend to suffer from start-up difficulties and benefit from the vintage effect reflects results of other studies (Booker et al., 2007; Sass, 2006). Despite the State's effort to adequately train and prepare new charters for the tasks and responsibilities ahead, policies and guidance from the state and local education agencies regarding charter school operations may be necessary prior to schools opening and in their first few years of operation to mitigate the impact of being a new school.

In addition, since charter schools, by definition, are enrolled in by choice, it is expected they have high

Table 7

Estimated effects of charter school attendance on student achievement: Hierarchical linear growth models based on “switchers”.

	Math grades 1–6				Language arts grades 1–6				Language arts grades 7–11			
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)
Growth rate	.001 (.002)	–.001 (.003)	–.001 (.003)	–.001 (.003)	.0003 (.002)	–.002 (.002)	–.002 (.002)	–.002 (.002)	–.004** (.001)	–.018** (.001)	–.018** (.001)	–.018** (.001)
Charter school	–.032** (.005)	–.044** (.006)			–.056** (.005)	–.088** (.005)			–.001 (.006)	.005 (.007)		
1st year			–.145** (.008)	–.112** (.012)			–.139** (.008)	–.137** (.011)			–.059** (.012)	–.044** (.016)
2nd year			–.063** (.008)	–.044** (.009)			–.108** (.008)	–.104** (.009)			.011 (.011)	.024 (.013)
3rd year			–.016 (.009)	.0004 (.009)			–.074** (.008)	–.071** (.009)			–.011 (.011)	0 (.013)
4th year			.021 (.011)	.037** (.011)			–.057** (.01)	–.054** (.011)			.036** (.013)	.045** (.014)
5th year			.126** (.014)	.142** (.014)			.026* (.013)	.029* (.013)			.053** (.016)	.063** (.017)
6th year/more			.107** (.015)	.123** (.015)			.008 (.014)	.012 (.014)			.038* (.017)	.046** (.017)
Mover		–.044** (.002)	–.039** (.002)			–.026** (.002)	–.023** (.002)			–.025** (.003)	–.024** (.003)	
TPS-to-TPS mover				–.04** (.002)				–.025** (.002)				–.075** (.003)
TPS-to-charter mover				–.073** (.01)				–.023* (.01)				–.086** (.011)
Charter-to-TPS mover				.068** (.014)				.029* (.013)				–.032** (.011)
Charter-to-charter mover				–.014 (.025)				–.004 (.023)				–.042 (.024)
Structural mover		–.004 (.005)	–.007 (.005)	–.005 (.005)		–.021** (.004)	–.022** (.004)	–.021** (.004)		–.016** (.003)	–.016** (.003)	–.017** (.003)
Student controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
School controls	No	Yes	Yes	Yes	No	Yes	Yes	Yes	No	Yes	Yes	Yes
Deviance	2,680,156	2,509,939	2,509,716	2,509,649	2,657,510	2,444,588	2,444,469		1,866,133	1,658,375		
# obs.			955,456				951,383				680,202	
# students			230,808				230,808				177,400	

The estimates are based solely on a sample with “switchers”, or students who switched schools between charter schools and TPSs during 2004 and 2009. This is the same set of students who were included in the GMM estimations. See Table 5 for more notes.

student mobility, especially in their early operational years. Our analysis indicates that different types of student mobility in charters tend to have different effects on student achievement. In most of the estimated models, transfers between TPSs and between charters tend to have small negative or no effect on student achievement. By contrast, student transfers from TPSs to charter schools consistently show negative achievement gains, especially in elementary schools, while in other cases the transfers from charter schools to TPSs actually have positive effects on student achievement. This raises questions regarding a charter schools' ability and effectiveness, as compared to TPSs, to integrate new students into the classroom culture and learning environment. Given the fact that newly established charter schools tend to have much higher student turnover rates than more established charter schools, part of the negative effect of charter schools in their early development may be attributed to this condition. It is necessary to provide extra support to better prepare them to integrate new students.

As charter schools in Utah are funded on the principle that state funds follow the student, the financial costs associated with the enrollment loss to charter schools are expected to generate competitive reactions from TPSs. This systemic effect is not estimated in this study. Depending on the direction of the effect, it may bias the direct effect of charter schools in different ways. Existing research on charter schools' systemic effects, however, remains at an early stage, indicating no consistent or substantial improvements or harms in public school systems (Ni & Arsen, 2010). As such, we expect little systemic effect of Utah charter schools on TPSs in general, as the charter enrollment only accounted for 5% of total public school enrollment in the state. As a fast and consistently growing state, it is less likely for Utah TPSs to feel much competitive pressure from charter schools. On the other hand, the charter enrollment is not evenly distributed among districts in the state. It is reasonable to expect more acute pressure in some school districts than others. As charter school enrollment continues to expand at a rapid pace, we anticipate more systemic reactions from TPSs. These issues will be part of our future studies.

In addition, our analysis focuses only on the average charter school effect on student achievement. Yet, we recognize that the mission of charter schools may be more heterogeneous than TPSs. That is, many have a specialized focus area. Future research is needed to examine the distribution of effective charter schools, including by mission, and to identify the most effective ones and their successful practices, particularly those charters that are able to mitigate the negative effects often associated with early start-up and mobility.

Finally, we have only considered performance on state administered CRTs here; other academic and non-academic outcomes need further attention in our research. For example, many parents in Utah choose charter schools not because of their overall academic performance. Instead, they choose charters because of issues related to their child's anticipated academic success and safety, perceived curriculum and instructional innovation, access to decision-making and control, attending schools with

children more similar to the characteristics of their child, and convenience (Rorrer et al., 2006). Thus, even if charter schools do not perform better than TPSs, they may still be perceived as a more attractive choice if students are better matched with a particular charter school mission than their assigned public schools.

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